

Selvam College of Technology (Autonomous),

"A" Grade by NAAC, UGC recognized 2(f) Status, Approved by AICTE – New Delhi, Affiliated to Anna University

Namakkal – 03. www.selvamtech.edu.in

DEPARTMENT OF INFORMATION TECHNOLOGY

SYLLABUS

IT3501

FULL STACK WEB DEVELOPMENT

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COURSE OBJECTIVES:

- To understand the various components of full stack development
- To learn Node.js features and applications
- To develop applications with MongoDB
- To understand the role of Angular and Express in web applications
- To develop simple web applications with React

UNIT I

BASICS OF FULL STACK

9

Understanding the Basic Web Development Framework - User - Browser – Webserver - Backend Services – MVC Architecture - Understanding the different stacks –The role of Express – Angular – Node – Mongo DB – React .

UNIT II

NODE JS

9

Basics of Node JS – Installation – Working with Node packages – Using Node package manager – Creating a simple Node.js application – Using Events – Listeners –Timers - Callbacks – Handling Data I/O – Implementing HTTP services in Node.j.

UNIT III

MONGO DB

9

Understanding NoSQL and MongoDB – Building MongoDB Environment – User accounts – Access control – Administering databases – Managing collections – Connecting to MongoDB from Node.js – simple applications.

UNIT IV

EXPRESS AND ANGULAR

9

Implementing Express in Node.js - Configuring routes - Using Request and Response objects - Angular - Typescript - Angular Components - Expressions - Data binding - Built-in directives.

UNIT V

REACT

9

MERN STACK – Basic React applications – React Components – React State – Express REST APIsModularization and Webpack - Routing with React Router – Server-side rendering

TOTAL:45 PERIODS

COURSE OUTCOMES:

- CO1: Understand the various stacks available for web application development
CO2: Use Node.js for application development
CO3: Develop applications with MongoDB
CO4: Use the features of Angular and Express
CO5: Develop React applications

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TEXT BOOK :

1. Brad Dayley, Brendan Dayley, Caleb Dayley, ‘Node.js, MongoDB and Angular Web Development’, Addison-Wesley, Second Edition, 2018
2. Vasan Subramanian, ‘Pro MERN Stack, Full Stack Web App Development with Mongo, Express, React, and Node’, Second Edition, Apress, 2019.

REFERENCES :

1. Chris Northwood, ‘The Full Stack Developer: Your Essential Guide to the Everyday Skills Expected of a Modern Full Stack Web Developer’, Apress; 1st edition, 2018
2. Kirupa Chinnathambi, ‘Learning React: A Hands-On Guide to Building Web Applications Using React and Redux’, Addison-Wesley Professional, 2nd edition, 2018
3. https://www.tutorialspoint.com/the_full_stack_web_development/index.asp
4. <https://www.coursera.org/specializations/full-stack-react>
5. <https://www.udemy.com/course/the-full-stack-web-development/>